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A1) Probability of success in any given trial is
 $p = 1/4$ $\therefore$ Binomial Distribution

$Y \sim \text{Bin}(n, p)$ $\{n: \text{number of trials}, p: \text{Prob succ}\}$

asked to find smallest value of N such that
Probability of at least one success is greater
than or equal to 0.7

$$\hookrightarrow P(Y \geq 1) \geq 0.7$$

$$P(Y \geq 1) = 1 - P(Y = 0)$$

$$1 - P(Y = 0) \geq 0.7$$

for Binomial dist, Probability of getting
exactly k successes in n trials is given
by

$$P(Y = k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

$$k=0, p=1/4, 1-p=3/4$$

$$P(Y=0) = \binom{n}{0} \cdot \left(\frac{1}{4}\right)^0 \cdot \left(\frac{3}{4}\right)^n = 1 \cdot 1 \cdot \left(\frac{3}{4}\right)^n$$

$$\therefore 1 - \left(\frac{3}{4}\right)^n \geq 0.7$$

$$\left(\frac{3}{4}\right)^n \leq 0.3$$

$$n \cdot \ln\left(\frac{3}{4}\right) \leq \ln(0.3)$$

-ve value

$$n \geq \frac{\ln(0.3)}{\ln(3/4)}$$

$$n \geq 4.185$$

$$\therefore n = 5 \text{ smallest}$$

A2) given

$$X \sim \text{Bin}(2, p) \text{ and } P(X \geq 1) = 5/9$$

$$P(X \geq 1) = 1 - P(X = 0) = \frac{5}{9}$$

$$P(X = 0) = \frac{4}{9}$$

$$\frac{4}{9} = \binom{n}{k} p^k (1-p)^{n-k} = \text{Bin}(2, p)$$

$$\frac{4}{9} = \binom{2}{0} p^0 (1-p)^{2-0}$$

$$\frac{4}{9} = 1 \cdot 1 \cdot (1-p)^2 \quad 1-p = \pm \frac{2}{3}$$

$$P = 5/3$$



not possible.

greater than 1

$$\Rightarrow P(Y \geq 1)$$

$$Y \sim \text{Bin}(4, P)$$

$$P = 1/3$$

$$P(Y \geq 1)$$

$$P(Y \geq 1) = 1 - P(Y = 0)$$

$$Y \sim \text{Bin}(4, 1/3), \quad 1 - P = 2/3$$

$$P(Y = 0) = \binom{4}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{4-0} = \frac{16}{81}$$

$$P(Y \geq 1) = 1 - \frac{16}{81} = \frac{65}{81}$$

A3)

Definition of a unique mode

Mode of a discrete probability distribution is the value for which the PMF is at its maximum. If $x = a$ is the unique mode for the random variable X , that it means probability at ' a ' is strictly greater than the probability at its neighboring value $a-1$ and $a+1$.

$$\begin{aligned} \Rightarrow P(X=a) &> P(X=a-1) \\ P(X=a) &> P(X=a+1) \end{aligned}$$

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\binom{n}{a} p^a (1-p)^{n-a} > \binom{n}{a-1} p^{a-1} (1-p)^{n-a+1}$$

$$\frac{n!}{(n-a)! a!} p^a (1-p)^{n-a} > \frac{n!}{(n-a+1)! (a-1)!} p^{a-1} (1-p)^{n-a+1}$$

$$\frac{p}{(n-a+1) \cdot a!} > \frac{(1-p)}{(n-a+1)! \cdot (a-1)!}$$

$$\frac{p}{a} > \frac{1-p}{n-a+1}$$

$$pn - pa + p > a - ap$$

$$np + p > a$$

from 2nd Inegu

$$\Rightarrow P(X=a) > P(X=a+1)$$

$$\frac{p^a (1-p)^{n-a}}{(n-a)! a!} > \frac{p^{a+1} (1-p)^{n-a-1}}{(a+1)! (n-a-1)!}$$

$$\frac{1-P}{n-q} > \frac{P}{q+1}$$

$$q + 1 - qP - P > nP - qP$$

$$q > nP + P - 1$$

$$q > P(n+1) - 1 \quad \therefore \text{from 2nd Ineq}$$

Combining results.

$$q > (n+1)P - 1$$

$$q < (n+1)P$$

$\therefore$ yields

$$(n+1)P - 1 < q < (n+1)P$$

A4)

Step 1 - Parameter λ

$X \sim \text{Poisson}(\lambda)$

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

k : number of events

given condition $P(X=1) = P(X=2)$

$$\frac{e^{-\lambda} \cdot \lambda^1}{1!} = \frac{e^{-\lambda} \cdot \lambda^2}{2!}$$

$$\frac{2!}{1!} = \lambda \Rightarrow \lambda = 2$$

$$P(X=3)$$

$$= \frac{e^{-2} \cdot 2^3}{3!} = \frac{e^{-2} \cdot (2)^3}{3!}$$

$$P(X=3) = \frac{e^{-2} \cdot 8}{6} = \frac{4e^{-2}}{3}$$

A5) Let X number of geese in Pond &
 $X \sim \text{Poisson}(\lambda)$ & mean 2

$$P(X=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$$P(X \geq 2) > 0.99$$

$$P(X \geq 2) = 1 - [P(X=0) + P(X=1)] > 0.99$$

$$P(X=0) + P(X=1) < 0.01$$

$$0.01 > e^{-\lambda} + \lambda e^{-\lambda}$$

$$0.01 > e^{-\lambda} (1 + \lambda)$$

Using calc $\lambda \approx 6.63$

Smallest value λ mean can take is 6.63

A6) a) factorial moment $E[X(X-1)]$ for Binomial dist

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ for } k=0, 1, \dots, n$$

second factorial moment is calculated using

$$E[X(X-1)] = \sum_{k=0}^n k(k-1) \binom{n}{k} p^k (1-p)^{n-k}$$

$k=0, k=1$ and 2 up, $k=2$

$$= \sum_{k=2}^n \frac{n!}{(k-2)! (n-k)!} p^k (1-p)^{n-k}$$

Simplify the coefficient by expanding the factorial

$$E[X(X-1)] = \sum_{k=2}^n \frac{n!}{(k-2)! (n-k)!} p^k (1-p)^{n-k}$$

$$E[X(X-1)] = \sum_{k=2}^n n(n-1) \binom{n-2}{k-2} p^k (1-p)^{n-k}$$

$$= n(n-1) p^2 \sum_{k=2}^n \binom{n-2}{k-2} p^{k-2} (1-p)^{n-k}$$

$$j = k - 2$$

$$E[X(X-1)] = n(n-1) p^2 \sum_{j=0}^{n-2} \binom{n-2}{j} p^j (1-p)^{(n-2)-j}$$

summation term is $(p + (1-p))^{n-2} = 1$

$$\therefore E[X(X-1)] = n(n-1) p^2 (1) = n(n-1) p^2$$

A(b)

$E[X(X-1)]$ for Poisson distribution is

$$P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}, \text{ for } k = 0, 1, 2, 3, \dots$$

Using definition of expected value:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1) P(X=k) = \sum_{k=0}^{\infty} k(k-1) \frac{e^{-\lambda} \lambda^k}{k!}$$

Since term for $k=0, k=1$ are zero
we can start summation from $k=2$:

$$E[X(X-1)] = \sum_{k=2}^{\infty} k(k-1) \frac{e^{-\lambda} \lambda^k}{k!}$$

$$k! = k(k-1)(k-2)!$$

$$E[X(X-1)] = \sum_{k=2}^{\infty} \frac{e^{-\lambda} \lambda^k}{(k-2)!}$$

Factor out constant $e^{-\lambda} \lambda^2$

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!}$$

$$\text{Let } j = k-2$$

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \left(\sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \right)$$

A 6 (c)

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[X(X-1)] = E[X^2 - X]$$

by Linearity

$$= E[X^2] - E[X]$$

$$E[X^2] = E[X(X-1)] + E[X]$$

$$\text{Var}(X) = (E[X(X-1)] + E[X]) - (E[X])^2$$

$$X \sim \text{Bin}(n, p)$$

$$E[X] = np$$

$$E[X(X-1)] = n(n-1)p^2$$

$$\text{Var}(X) = (n(n-1)p^2 + np) - (np)^2$$

$$= (n^2p^2 - np^2 + np) - n^2p^2$$

$$= n^2p^2 - np^2 + np - n^2p^2$$

$$\text{Var}(X) = np - np^2$$

$$\boxed{\text{Var}(X) = np(1-p)}$$

Ab (d)

A7) X be the number of errors the chicken
sex makes

a) Exact Probability using Bino Distribution

1000 independent trials $n=1000$

error / no error

$$n=1000$$

$$\text{Probability of success (error)} p = \frac{1}{500} = 0.002$$

$$X \sim \text{Bin}(1000, 0.002)$$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$\binom{1000}{0} (0.002)^0 (0.998)^{1000} \approx 0.135$$

$$\binom{1000}{1} (0.002)^1 (0.998)^{999} \approx 0.270$$

$$\binom{1000}{2} (0.002)^2 (0.998)^{998} \approx 0.270$$

Summing them

$$\boxed{\sim 0.676}$$

b) Poisson distribution can be used to
approximate Binomial dist when n is
large and probability of success is
small.

$$\text{Mean } \lambda = nP = 1000 \cdot (0.002) = 2$$

We can approximate X with new RV $Y \sim \text{Poisson}(2)$

$$\begin{aligned} P(Y \leq 2) &= P(Y=0) + P(Y=1) + P(Y=2) \\ \lambda=2 &= \frac{e^{-2} \cdot 2^0}{0!} + \frac{e^{-2} \cdot 2^1}{1!} + \frac{e^{-2} \cdot 2^2}{2!} \end{aligned}$$

$$= 1 \cdot e^{-2} + 2e^{-2} + 2e^{-2} = 5e^{-2} \approx 0.135 \times 5$$

$$P(Y \leq 2) = 0.67667$$

It is Very accurate in this case

A8) a) Selection ~~with~~ replacement (Binomial)

$$n=14$$

$$\text{Total Students} = 30$$

Students wearing wrist watches = 3

$$\text{Probability of success } P = \frac{3}{30} = \frac{1}{10} = 0.1$$

$$\text{Probability of failure} = 1 - P$$

$$= 1 - 0.1 = 0.9$$

$$P(X=2) = \binom{14}{2} (0.1)^2 (0.9)^{12}$$

$$\approx 0.2570$$

b) Selection without Replacement { Hypergeometric distribution }

$$N = 30$$

Success in Population $K = 3$

Sample size $n = 14$

Successes in sample $k = 2$

12 Student without watches

students without watches in Population

$$N - K = 30 - 3 = 27$$

Hypergeometric Probability formula

$$P(X=K) = \frac{\binom{K}{n} \binom{N-K}{n-K}}{\binom{N}{n}}$$

$$P(X=2) = \frac{\binom{3}{2} \binom{27}{12}}{\binom{30}{14}}$$

Using Calculator

$$\approx 0.35$$

A9) Let X be the number of weeks it takes to find an acceptable apartment.

P of success in any given week $P = 0.1$
 P of failure $= 0.9$

$$X \sim \text{Geo}(0.1)$$

a)

$$E[X] = \frac{1}{0.1} = 10$$

$$b) P(X \leq 2) = P(X=1) + P(X=2)$$

$$\text{PMF of geometric dist} = (1-P)^{k-1} P$$

$$P(X=1) = (0.9)^{1-1} (0.1) = 0.1$$

$$P(X=2) = (0.9)^{2-1} (0.1) = 0.09$$

$$P(X \leq 2) = 0.19$$

$$\boxed{19\%}$$

c) Number of weeks X is > 17

$$P(X > 17)$$

For it to take more than 17 weeks, the apartment seeker must fail to find an acceptable apartment for each of the first 17 weeks.

Since trials are independent we can multiply

their Probabilities

$$P(X > 17) = (\text{Prob of failure})^7 \\ = (1-p)^7 = (0.9)^7 \approx 0.4783$$

$$\boxed{16.67\%}$$

A10) a) Exact Probability.

Using hypergeometric distribution

$$N = 100$$

of successes in population $K = 25$

$n = 10$ { sample size }

$k = 2$ { desired success }

$$P(X=2) = \frac{\binom{25}{2} \cdot \binom{75}{8}}{\binom{100}{10}}$$

$$\approx \boxed{0.2935}$$

b) $N = 100$

$n = 10$

$$p = \frac{25}{100} = 0.25$$

$$P(X=2) = \binom{10}{2} (0.25)^2 (1-0.25)^{10-2}$$

$$P(X=2) \approx 0.2815$$

Ans) Step-2)

$$\text{Given } P(X=0) + P(X=1) = 1$$

Since this is a R.V. can only take 2 values it implies X can only take 0 or 1.
 $\therefore X$ is Bernoulli random variable.

$$P(X=1) = p$$

$$P(X=0) = 1-p \quad \text{in terms of } p$$

2) Exp & Var in terms of p

$$E[X] = \sum x_i P(X=x_i)$$

$$= (0 \cdot P(X=0)) + (1 \cdot P(X=1))$$

$$= (0 \cdot (1-p)) + (1 \cdot p) = p$$

Var

$$E[X^2] = \sum x_i^2 P(X=x_i)$$

$$= (0^2 \cdot P(X=0)) + (1^2 \cdot P(X=1))$$

$$= (0 \cdot (1-p)) + (1 \cdot p) = p$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2$$

$$= p(1-p)$$

Step - 3

$$P = 3P - 2P^2$$

$$3P^2 + P - 3P = 0$$

$$3P^2 - 2P = 0$$

$$P(3P - 2) = 0$$

$$P = 0$$

or

$$P = 2/3$$

if $P=0$ $E[X]=0$ and $\text{var}[X]=0$ satisfies
if ~~$P=1$~~ but degenerate solution outcome
always 0.

$P = 2/3$ non trivial r.v

Step-4

using $P = 2/3$

$$P(X=1) = P = 2/3$$

$$P(X=2) = 1 - P(X=1) = 1 - 2/3 = \boxed{1/3}$$